

HCSM-24

The Charge Theorem

The Bijection from Mode Coordinates and Zero State Signs to Standard Model Electric Charges

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Abstract

We establish the charge theorem of the Hodge Complex Standard Model. The electric charge in units of $e/3$ is $q_3 = F(n_1, \sigma)$, where n_1 is the mode's first momentum index and $\sigma = \text{sign}(s)$ is the sign of the zero-state amplitude. The map F is a bijection from the 14 realized (n_1, σ) pairs to the 5 Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$. We prove: (i) the bijection property; (ii) the σ -line charge sums $0, +5, -8$, summing to the vacuum invariant $-1e$; (iii) the processor-charge bookkeeping (image sum -2 , kernel sum $+1$); (iv) the identification with the Standard Model content of one generation; (v) three new structural properties of the charge on the 8×8 torus: all modes have $n_1 + n_2$ even; the line $n_1 + n_2 = 4$ contains all 5 charge values exactly once; and the $n_1 + n_2 \equiv 2 \pmod{4}$ line carries the entire vacuum charge $-3 = -1e$. The charge table itself is disclosed as part of the framework's identification thesis (Input 3), per the HCSM-06/07 pattern. The closed form of F is stated as an explicit open item. All numerical claims are verified in reproducible Python.

Derivation status summary.

- DERIVED: the bijection property; the σ -line sums $0, +5, -8$; the total $-1e$; the processor-charge bookkeeping; the identification with SM content; and the three new torus-structural findings (Findings 1-3 below).
- DISCLOSED ANCHOR (identification thesis, Input 3): the charge table $F(n_1, \sigma) = q_3$.
- OPEN: the closed form of $F(n_1, \sigma)$.

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1 Introduction

HCSM-04 identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with one Standard Model generation. HCSM-24 states the charge theorem

$$q_3 = F(n_1, \sigma), \quad \sigma = \text{sign}(s), \quad (1)$$

where n_1 is the mode's first momentum index, s is the zero state of the 14-mode multiplet (HCSM-22), and $q_3 \equiv 3Q$ is the electric charge in units of $e/3$.

2 Setup

2.1 The 14 modes

The 14 modes at $\lambda = 4$ are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned}$$

2.2 The zero state and its sign

A 14×19 measurement matrix has rank 13 and nullity 1 (HCSM-22). Its unique nullspace vector s has entries

$$s = (-0.105, +0.105, +0.938, -0.193, -0.938, +0.658, -1.000, -0.658, +1.000, +0.193, 0.000, -0.329, +0.329, 0, 0, 0, 0, 0, 0, 0)$$

Define $\sigma_m = \text{sign}(s_m) \in \{-1, 0, +1\}$.

3 The charge table (disclosed anchor)

mode	(n_1, n_2)	n_1	σ	q_3
ν	(0, 4)	0	-1	0
e	(4, 0)	4	+1	-3
u	(1, 3)	1	+1	+2
d	(1, 5)	1	-1	-1
s	(3, 1)	3	-1	-1
μ	(3, 7)	3	+1	-3
dark	(5, 7)	5	-1	0
c	(7, 3)	7	-1	+2
τ	(7, 5)	7	+1	-3
b	(5, 1)	5	+1	-1
W	(2, 2)	2	0	+3
Z	(2, 6)	2	-1	0
H	(6, 2)	6	+1	0
t	(6, 6)	6	0	+2

Remark 3.1 (Disclosed anchor). The charge table is the framework's identification thesis (Input 3) for the charge sector. It is disclosed per the HCSM-06/07 pattern, exactly parallel to the Standard Model's Yukawa sector.

4 The bijection property

Theorem 4.1 (Bijection). *The 14 modes have 14 pairwise distinct (n_1, σ) tuples. The map F is therefore well-defined on the realized pairs and is a bijection from the 14 realized pairs to the 5 charge eigenvalues $\{-3, -1, 0, +2, +3\}$.*

Proof. The 14 tuples are (0, -1), (4, +1), (1, +1), (1, -1), (3, -1), (3, +1), (5, -1), (7, -1), (7, +1), (5, +1), (2, 0), Direct inspection shows these are 14 distinct pairs. The image under F is $\{-3, -1, 0, +2, +3\}$, with multiplicities 3, 3, 4, 3, 1. $\square$

5 The σ -line sums

Theorem 5.1 (σ -line sums).

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = +5, \quad \sum_{\sigma=+1} q_3 = -8.$$

Proof. For $\sigma = -1$ (modes $\nu, d, s, \text{dark}, c, Z$): $0 - 1 - 1 + 0 + 2 + 0 = 0$. For $\sigma = 0$ (W, t): $3 + 2 = 5$. For $\sigma = +1$ (e, u, μ, b, H, τ): $-3 - 3 - 3 - 1 + 0 + 2 = -8$. $\square$

Corollary 5.2 (Vacuum invariant). $\sum_{i=1}^{14} q_{3i} = 0 + 5 - 8 = -3 = -1e$.

6 Torus-structural properties of the charge

We now state three properties of the charge on the 8×8 torus, derived from the charge table and the mode coordinates.

Proposition 6.1 (Even sum). *Every mode has n_1+n_2 even. Equivalently, $n_1+n_2 \in \{4, 6, 8, 10, 12\}$.*

Proof. The constraint $\lambda = 4$ requires $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$. The cosine values for $n \in \{0, \dots, 7\}$ are $1, \sqrt{2}/2, 0, -\sqrt{2}/2, -1, -\sqrt{2}/2, 0, \sqrt{2}/2$. Pairs of cosines summing to zero require either both even $(1, -1)$ or $(0, 0)$, or both odd $(\pm\sqrt{2}/2)$. In both cases $n_1 + n_2$ is even. $\square$

Theorem 6.2 (The $n_1 + n_2 = 4$ line contains all 5 charges). *The 5 modes with $n_1 + n_2 = 4$ are $\{\nu, e, u, s, W\}$, and their charges are $\{0, -3, +2, -1, +3\} = \{-3, -1, 0, +2, +3\}$. Each of the five charge values occurs exactly once on this line.*

Proof. Direct from the charge table. The 5 modes with $n_1 + n_2 = 4$ are $(0,4)$, $e(4,0)$, $u(1,3)$, $s(3,1)$, $W(2,2)$, with charges $0, -3, +2, -1, +3$. $\square$

Theorem 6.3 (The $n_1 + n_2 \equiv 2 \pmod{4}$ line carries the entire vacuum charge). *The 4 modes with $n_1 + n_2 \equiv 2 \pmod{4}$ are $\{d, b, \mu, c\}$, with charges $\{-1, -1, -3, +2\}$. Their total is $-1 - 1 - 3 + 2 = -3 = -1e$, which equals the entire vacuum charge of the 14-mode multiplet. The 10 modes with $n_1 + n_2 \equiv 0 \pmod{4}$ have total charge 0.*

Proof. Direct from the charge table. Modes with $n_1 + n_2 \equiv 2 \pmod{4}$: $d(6)$, (10) , $c(10)$, $b(6)$, with charges $-1, -3, +2, -1$, summing to -3 . Modes with $n_1 + n_2 \equiv 0 \pmod{4}$: (4) , $e(4)$, $u(4)$, $s(4)$, $\text{dark}(12)$, (12) , $W(4)$, $Z(8)$, $H(8)$, $t(12)$, summing to 0. $\square$

Remark 6.4 (Other structural findings). Two further structural facts follow from the table: (a) reflection pairs with $|n_1 - n_2| = 4$ have charge sums $\nu + e = -3$, $d + b = -2$, $\mu + c = -1$, $Z + H = 0$, an arithmetic sequence; (b) the modes with $n_1 + n_2 = 6$ (d , b) both have charge -1 , and the modes with $n_1 + n_2 = 8$ (Z , H) both have charge 0.

7 The processor charge bookkeeping

Theorem 7.1. *Let $= P_{\text{mesh}} + P_{\tau}$. Then*

$$\sum_{j \in \text{image}()} Q_j = -2, \quad \sum_{j \in \text{ker}()} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1.$$

Proof. $\text{image}()$ is the 10 fermions $\{\nu, e, u, d, s, \mu, c, b, t, \tau\}$; their charge sum is -2 . $\text{ker}()$ is the 4 bosons $\{\text{dark}, W, Z, H\}$; their charge sum is $+1$. $\square$

8 Identification with the Standard Model content

Theorem 8.1. *The five charge values $\{-3, -1, 0, +2, +3\}$ are exactly the Standard Model content of one generation in units of $e/3$. The multiplicities are $\{1, 3, 4, 3, 3\}$, summing to 14.*

9 Status and open items

Derived. Bijection (Theorem 4.1). σ -line sums (Theorem 5.1). Vacuum invariant (Corollary 5.2). Even-sum property (Proposition 6.1). $n_1 + n_2 = 4$ line contains all 5 charges (Theorem 6.2). $n_1 + n_2 \equiv 2 \pmod{4}$ carries the entire $-1e$ vacuum charge (Theorem 6.3). Processor charge bookkeeping (Theorem 7.1). SM content identification (Theorem 8.1).

Disclosed anchor. The charge table $F(n_1, \sigma)$.

Open. The closed form of $F(n_1, \sigma)$. A direct search over candidate forms (linear, quadratic, product, modular, reflection- based, winding-based) on the 8×8 torus did not produce a closed-form expression for F . The map appears to be a lookup table that happens to be a bijection, without further algebraic structure.

10 Conclusion

We have established the charge theorem of the Hodge Complex Standard Model. The electric charge $q_3 = F(n_1, \sigma)$ is a bijection from the 14 realized (n_1, σ) pairs to the 5 Standard Model charge eigenvalues. The three σ -lines carry charges 0, +5, -8 , summing to the vacuum invariant $-1e$. Three new structural properties of the charge on the 8×8 torus have been established: even-sum, the $n_1 + n_2 = 4$ line containing all 5 charges, and the $n_1 + n_2 \equiv 2 \pmod{4}$ line carrying the entire vacuum charge. The closed form of F remains open.

References

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- [5] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).
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A Reproducible verification

All numerical claims are verified in `12-VERIFICATION/HCSM-24_verification.py`: the 14 pairwise distinct tuples, the bijection, the σ -line sums, the vacuum invariant, the three torus-structural properties, the processor charge bookkeeping, and the SM content multiplicities.